

FITBUDDY - AI FITNESS PLAN GENERATOR USING GEMINI MODELS

Phase 3: Project Design Phase

This phase focuses on designing the overall architecture, database, AI integration, backend, and frontend structure.

System Architecture

FitBuddy follows a modular architecture consisting of:

Frontend → FastAPI Backend → AI Layer / Database

The major components are:

- **Frontend:** HTML, CSS, and Jinja2 templates.
- **Backend:** FastAPI for routing and request processing.
- **AI Layer:** Google Gemini API.
- **Database:** SQLite with SQLAlchemy ORM.
- **Server:** Uvicorn.

Main Application Components

- `main.py` – Application entry point.
- `routes.py` – Routing and application logic.
- `gemini_generator.py` – Workout generation.
- `gemini_flash_generator.py` – Nutrition/recovery tips.
- `updated_plan.py` – Feedback-based plan modification.
- `database.py` – Database operations.

- `templates/` – Jinja2 HTML templates.
- `static/` – CSS and image assets.

Frontend Design

Three major templates are designed:

- `index.html` – User input page.
- `result.html` – Workout plan and nutrition tip page.
- `all_users.html` – Administrative user and plan view.

Database Design

SQLite and SQLAlchemy are used to store:

- User details.
- Original workout plans.
- Updated workout plans.
- Related fitness information.

Outcome:

A complete technical and UI architecture for the FitBuddy system.